

File Edit View Insert Runtime Tools Help

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✓ RAM
Disk

```
[16] ✓ 0s  
import pandas as pd  
from sklearn.preprocessing import LabelEncoder  
from sklearn.tree import DecisionTreeClassifier  
from sklearn import tree
```

```
[17] ✓ 0s  
data = pd.read_csv("/content/Decision_Tree.csv")  
data
```

▼

	Machine	Temperature	Vibration	Failure	
0	1	High	High	Yes	
1	2	High	Low	Yes	
2	3	Medium	High	Yes	
3	4	Low	Low	No	
4	5	Low	Medium	No	
5	6	Medium	Low	No	
6	7	High	Medium	Yes	
7	8	Low	High	No	

Next steps: [Generate code with data](#) [New interactive sheet](#)

```
label_encoder = LabelEncoder()
data["Temperature"] = label_encoder.fit_transform(data["Temperature"])
data["Vibration"] = label_encoder.fit_transform(data["Vibration"])
data["Failure"] = label_encoder.fit_transform(data["Failure"])
```

```
x = data[["Temperature", "Vibration"]]
y = data["Failure"]
print(x)
print(y)
```

	Temperature	Vibration
0	0	0
1	0	1
2	2	0
3	1	1
4	1	2
5	2	1
6	0	2
7	1	0
0	1	
1	1	
2	1	
3	0	
4	0	
5	0	
6	1	

```
[20] ✓ 0s model = DecisionTreeClassifier(criterion="entropy")
model.fit(x,y)

***
DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy')

[21] ✓ 0s sample=[[1,1]]

[22] ✓ 0s prediction=model.predict(sample)
print(prediction)

[0]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature
warnings.warn(

[23] ✓ 0s if prediction == 1:
    print("The machine will fail")
else:
    print("The machine will not fail")

The machine will not fail

[25] ✓ 0s import matplotlib.pyplot as plt
```



